Sector A branch-aware synthesis and termination package

TECT verification-first repository · claim A5-SECTOR-A-SYNTHESIS

first issued 2026-07-18 · this version issued 2026-07-19 · v1.2

1. Purpose and scope: result and current review boundary

This package answers one narrow completion question: do the frozen Sector-A records form a reproducible path from definitions through full variational dynamics and spectral PDE approximation, together with controlled scalar perturbative and non-perturbative continuum limits? The answer is

$$\boxed{\texttt{PASS@BRANCH-AWARE-DECLARED-SCOPE.}} \tag{1.1}$$

The qualifier is load-bearing. The records form two compatible but distinct branches, not one parameter-identical full constructive theory. The first is the full-production variational/PDE/discretization branch. The second is the positive-mass real-scalar perturbative/constructive branch.

Jusang Lee independently ran the integrated 32/32 command and approved v1.0 on 2026-07-19. That review enacted `T5 CLOSED@BRANCH-AWARE-SECTOR-A-SYNTHESIS`. It did not authorize a later, dependency-refreshed package automatically. Corrected A4 v2.1 is now operator-confirmed and PUBLISHED, so all six support bundles are complete. This v1.2 document is the exact capstone entry. After it was created and its initial PDF passed the form and overfull checks, Jusang Lee explicitly directed that the remaining approvals be processed in batch and that work continue. This records exact-package confirmation, conditional only on the unchanged source and final direct validation passing before bundle creation.

2 Common definitions and conventions

The common spatial domain is the fixed spectral torus

$$\mathbb{T}^3 = [0, L_x) \times [0, L_y) \times [0, L_z), \qquad L_x = L_y = L_z = 16. \tag{2.1}$$

For a three-component complex field Ψ the real pairing is

$$\langle u, v\rangle_{\mathbb{R}} = \mathrm{Re} \int_{\mathbb{T}^3} u^\dagger v \, dx. \tag{2.2}$$

The full-production fourth-order scalar symbol inside each component is

$$K_{\mathrm{full}}(k) = r + Z|k|^2 + Y|k|^4 = m_{\mathrm{full}}^2 + Y(|k|^2 - q_0^2)^2, \tag{2.3}$$

where $Y = 1$, $Z < 0$, $q_0^2 = -Z/(2Y)$, and

$$m_{\mathrm{full}}^2 = r - \frac{Z^2}{4Y}. \tag{2.4}$$

The scalar-continuum branch uses

$$K_{\mathrm{sc}}(k) = m_{\mathrm{sc}}^2 + Y(|k|^2 - q_0^2)^2 \tag{2.5}$$

with positive shell mass. Both branches use the spectral Fourier convention and $\eta_{\text{shell}} = 0$ in the closed statements.

The local polynomial convention shared by the verified scalar reduction and the scalar Gibbs construction is

$$V(\phi) = \int_{\mathbb{T}^3} \left(\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6 \right) dx, \qquad \gamma > 0. \tag{2.6}$$

The full P1 functional has more field structure: three complex components, family masses, an internal lock, positive regularisers, and derivative Class-II currents. Equation (2.6) is a verified scalar slice, not a statement that the full and scalar measures coincide.

3 Six frozen input results

D1: scalar kernel definition. `A1-PRODUCTION-KERNEL-MANIFEST` is T5. It pins the scalar Brazovskii symbol, Fourier convention, q_0, Y, Z, shell mass, and $\eta_{\text{shell}} = 0$. It is a definition and implementation-identity result, not a phase-selection theorem.

D2: full variational realization. `A1-PRODUCTION-FUNCTIONAL-REALISATION` is T5 within its declared finite-grid audit scope. On grids 4, 6, and 8, five field classes, and three finite-difference steps it verifies

$$DF[\Psi](v) = \langle R[\Psi], v\rangle_{\mathbb{R}}, \quad DR[\Psi]v = H[\Psi]v, \quad \langle u, Hv\rangle_{\mathbb{R}} = \langle Hu, v\rangle_{\mathbb{R}}. \tag{3.1}$$

It also verifies the local scalar reduction. This standalone canonical functional does not repair or certify the historical non-variational solver.

D3: full PDE theorem. `A2-FULL-PRODUCTION-WELLPOSED` is a T6 conditional theorem for the D2 functional on (2.1), with H^2 initial data, positive regularisers, and $\eta_{\text{shell}} = 0$. The linear operator is self-adjoint and bounded below; the coefficient conditions and positive sextic make the energy coercive; the gradient flow has a unique global H^2 solution, continuous finite-time dependence, exact energy dissipation, and smoothness at every positive time.

D4: spectral PDE approximation theorem. `A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM` is a T6 conditional theorem for the same D2/D3 branch. On pinned initial H^2 balls and $0 < \tau \le t \le T$, it provides finite explicit solution-ball and residual constants and proves restarted exact-Galerkin convergence. It covers positive time and exact nonlinear projection, not useful historical N32/N64/N128 error bars or a genuine finite-difference lattice.

D5: scalar perturbative cutoff removal. `A3-PERTURBATIVE-CONTINUUM-CORRELATORS` is a T6 conditional theorem. For the real scalar $d = 3$ positive-mass q^4 kernel, every connected perturbative amplitude converges at each fixed external momentum under the sharp spectral regulator. This is order-by-order, not a resummed construction.

D6: scalar constructive cutoff removal. `A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE` is a T6 conditional theorem. For $m_{\text{sh}}^2 > 0$, $Y > 0$, real λ, and $\gamma > 0$ on (2.1), the covariance is trace class, all sharp spectral Galerkin Gibbs measures are normalizable, and the full sequence converges weakly on real L^2. Lifted densities converge in total variation and the declared smeared cylinder- polynomial correlations converge. The corrected v2.1 proof isolates the zero Fourier mode when $q_0 = 0$ by using

$$m_0 = \max\Big(1, \left\lceil \sqrt{2}q_0/\alpha \right\rceil\Big), \tag{3.2}$$

so the nonzero max-shell count and inverse-power tail start at $m \ge 1$. This is finite volume and real scalar.

4 Branch graph and proof of its interfaces

The exact dependency graph is

```
shared spectral geometry and named hypotheses
  |
  +-- full-production branch
  |     A1-PRODUCTION-FUNCTIONAL-REALISATION
  |       -> A2-FULL-PRODUCTION-WELLPOSED
  |       -> A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
  |
  +-- scalar-continuum branch
        A1-PRODUCTION-KERNEL-MANIFEST
          +-> A3-PERTURBATIVE-CONTINUUM-CORRELATORS
          +-> A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
```

The two scalar arrows are parallel. They record a shared positive-mass spectral class, not a derivation of the A4 measure from a finite perturbation series and not a hard dependency between D5 and D6.

For the full branch, the A2 card names D2 as a hard dependency and carries `A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL`. The A3 full-discretization card names both D2 and D3 as hard dependencies and carries the same hypothesis. Its PDE residual, energy, solution ball, and exact-Galerkin flow therefore refer to the same canonical functional. This proves the two directed edges without identifying the historical solver.

For the scalar branch, D5 and D6 both state real-scalar, positive-mass, spectral/Galerkin scope. D5 uses q^{-4} domination at fixed external momenta. D6 uses the same covariance class with the local quartic/sextic potential and proves a non-perturbative finite-volume limit. D6 is general in positive mass and audits both production anchors without identifying them.

5 The mass fork is necessary

The canonical scalar kernel stores

$$m_{\mathrm{sc}}^2 = 0.005. \tag{5.1}$$

The full-production parameters store

$$r = 0.4740336473, \quad Z = -0.9252754126, \quad Y = 1,$$

so independent reconstruction gives

$$m_{\mathrm{full}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475. \tag{5.2}$$

Consequently

$$|m_{\mathrm{full}}^2 - m_{\mathrm{sc}}^2| = 0.255000000009475 > 0.2. \tag{5.3}$$

This is far above the upstream 10^{-9} kernel tolerance and is not rounding. A4 applies at either positive mass, but its generality does not identify the two anchored models. Any synthesis that writes (5.1)=(5.2) is false.

6 The functional fork is necessary

The D2 audit bounds the scalar-reduction relative error by

$$6.218564371786818 \times 10^{-16}. \tag{6.1}$$

This proves the local convention (2.6) when non-scalar structures are switched off. It does not prove equality of the full and scalar measures. The full Class-II density contains derivative currents. A4 Gaussian samples have covariance regularity only in H^s for $s < 1/2$; the derivative-current squares are not automatically defined there. Their construction requires a separate divergence and renormalisation analysis. No edge from D4 or D6 to a full three-component Class-II constructive measure is asserted.

7 Synthesis proposition and proof

Proposition. Under the six registered named hypotheses and exact scopes D1–D6, Sector A has: (i) a reproducible definitions-to-full-dynamics-to-spectral-approximation branch; (ii) a reproducible real-scalar spectral branch with both order-by-order and finite-volume non-perturbative cutoff removal; and (iii) an explicit interface contract that prevents parameter and functional identification across the branches.

Proof. D1 and D2 fix the scalar and full functional definitions. The hard-dependency edges in Section 4 compose D2, D3, and D4 into the full-production branch. D5 and D6 give distinct continuum statements on the scalar spectral class. Equations (2.1)–(2.6) verify shared geometry and local convention. Equations (5.1)–(5.3) prove that the anchored numerical models are not parameter-identical, while Section 6 proves that a scalar reduction is not a full-measure equivalence. The frozen manifest states these edges and non-edges. The primary and non-importing audits rehash all component records and support bundles and reconstruct every load-bearing interface. This proves the proposition at the declared synthesis scope. □

8 Content addressing and direct reproduction

The synthesis manifest pins six status cards, four component manifests, two component evidence records, and six PUBLISHED support-bundle manifests. Their content digests are

```
A1 scalar kernel:       99b255e767b5c1a6ea12983399f46ae5f23f5d3d76e4d587d3350c5c5ed62f7e
A1 full functional:    30749f1c5e1cdb9528ced2b6a3dcfced8d8ca8f6f39e8cef03913d32e13f33d6
A2 full PDE:           f07a39627a2eccc251fc67d1c988b9de18ec0b5643664fc60c3da0acc2eeeddb
A3 discretization:      6d15d165a73d3a2af07e10fce07394ce8b83311e571ba2aae2fbbc61c31d2e41
A3 scalar perturbative: 6783ee6637936675af9f0b16ede28fa5da91c1daf5c075e43f510625c59b9c0c
A4 scalar constructive: b1a215465956443ce22a7dcf42caaa9a3dfb61305759f4be4f55eab630cd3162
```

The primary audit checks every listed support file, recomputed digest, PUBLISHED marker, and runlog. The independent audit recomputes the six manifest digests, graph, Decimal mass fork, and exact capstone candidate hashes without importing the primary implementation.

Run from the repository root:

```
python codes/foundations/a5_sector_a_synthesis_verify.py
```

The required output is

```
PASS: primary (16/16)
PASS: independent (16/16)
ASSERTS: 32/32
A5-SECTOR-A-SYNTHESIS-INTEGRATED-PASS
Termination: PASS@BRANCH-AWARE-DECLARED-SCOPE
```

The wrapper executes both audits in a temporary directory, compares their branch and mass reconstructions, and writes the integrated result to the declared evidence path. The six support bundles remain the authoritative component reproduction packages; A5 verifies their current attestations.

9 What the termination result means

The result closes a documentation, interface, and reproducibility weakness in Sector A. The repository can answer which mathematical model is defined, which full field equation is controlled, why its spectral approximation converges in the pinned positive-time scope, and which scalar continuum limits exist perturbatively and non-perturbatively. It also records exactly where those statements stop.

This is not a claim that energy underwent a phase transition and created the present universe. No thermodynamic phase transition, vacuum selection, BCC existence, cosmology, or observation is proved. The result remains a scoped T5 synthesis closure, not a new T6 theorem and not T7 physical-domain closure.

10 Devil's-advocate review

1. **The P1–P4 sequence is one linear theory. UPHELD as false.** Sections 5 and 6 force the two-branch map.
2. **The two shell masses differ only by rounding. DISMISSED.** Their separation exceeds 0.2, versus a 10^{-9} upstream tolerance.
3. **A4 constructs the full derivative Class-II measure. UPHELD as false.** D6 is a real-scalar local-polynomial construction; the derivative currents need a separate analysis.
4. **The P1 scalar reduction proves measure equivalence. UPHELD as false.** It proves a local slice identity, not equality after restoring internal fields and derivative currents.
5. **Component T6 labels automatically make A5 T6. UPHELD as false.** A5 is a meta-level synthesis and interface audit; it adds no new conditional theorem beyond D1–D6.
6. **Six PUBLISHED support bundles make A5 unnecessary. DISMISSED.** No component package states or audits the cross-branch mass and functional non-implications.
7. **The graph secretly depends on the historical solver. DISMISSED in scope.** D2 is the standalone canonical functional; the historical non-variational source remains excluded.
8. **P3 gives useful error bars for historical grid runs. UPHELD as false.** D4 is an exact-Galerkin positive-time theorem with conservative constants and a distinct solver scope.
9. **Finite-volume cutoff removal proves a phase transition. UPHELD as false.** D6 contains no thermodynamic limit or spontaneous symmetry-breaking statement.
10. **This package closes BCC selection or Sector B. UPHELD as false.** Both are outside the proposition.
11. **The earlier v1.0 approval automatically authorizes v1.2. UPHELD as false.** v1.2 changes dependency pins, the A4 endpoint record, and the exact publication entry. Its authority instead comes from the later explicit 2026-07-19 batch-approval instruction, recorded here.
12. **The A4 q0=0 repair changes the A5 theorem. DISMISSED.** It repairs the endpoint notation and executable boundary inside D6; A5's branch proposition and T5 tier remain unchanged.
13. **A synthesis can omit a falsifier because it adds no physics. UPHELD as false.** Hash drift, a broken edge, or an erased fork falsifies the declared synthesis.

11 Falsifier and no-overclaim boundary

The synthesis is falsified by any failed source or evidence hash; a support bundle digest, file, marker, or runlog failure; a component no longer ACTIVE and reproducible in its pinned scope; an unregistered hypothesis; a missing D2 to D3 to D4 hard-dependency edge; a failed 32-assertion verifier; equality or silent substitution of (5.1) and (5.2); or a valid proof that D6 already defines the full derivative Class-II measure contrary to its card.

The package does not cover the historical non-variational solver, a full three-component derivative Class-II constructive measure, $\eta_{\text{shell}} \neq 0$, removal of the ρ or Class-II mass regularisers, $t = 0$ algebraic P3 rates from merely H^2 data, practical historical-grid error bars, finite-difference Route B, infinite volume, phase transition, minimizer uniqueness, BCC existence or selection, Sector B, physical-domain closure, T6, or T7.

12 Review and packaging order

The original A5 T5 review gate is CLOSED by the independently executed v1.0 approval. The dependency-first prerequisite is also SATISFIED: corrected A4 v2.1 is operator-confirmed and its PUBLISHED support bundle passes standalone 33/33, all file hashes, and its content digest. The exact v1.2 package review is CLOSED by the operator's explicit 2026-07-19 batch-approval instruction. The A5 PUBLISHED capstone bundle may therefore be built only from this unchanged source and validated PDF, then re-executed from its own root, integrity-checked, registered, and committed.

Result footer

```
Result ID:               A5-SECTOR-A-SYNTHESIS
Precise statement:       Sector A has a reproducible full-production
                         variational/PDE/Galerkin branch and a separate
                         real-scalar perturbative/constructive branch,
                         with exact interfaces and non-implications frozen.
Scope:                   T5 CLOSED@BRANCH-AWARE-SECTOR-A-SYNTHESIS;
                         fixed spectral T3, eta_shell=0; mass and
                         functional forks retained.
Dependencies:            Six component claims and their six PUBLISHED
                         support bundles; six named hypotheses.
Evidence grade:          EXACT + EXECUTED + CONDITIONAL.
Reproduction command:    python codes/foundations/
                         a5_sector_a_synthesis_verify.py
Expected output:         primary 16/16; independent 16/16; ASSERTS 32/32;
                         A5-SECTOR-A-SYNTHESIS-INTEGRATED-PASS;
                         PASS@BRANCH-AWARE-DECLARED-SCOPE.
Falsification gate:      Hash/digest/runlog drift, broken dependency or
                         hypothesis, failed audit, or erased branch fork.
Tier before / after:     T4 referee candidate / T5 after independent
                         operator execution and explicit v1.0 confirmation.
No-overclaim statement:  Not one parameter-identical full constructive
                         theory; not full Class-II measure, Route B,
                         infinite volume, phase transition, BCC, Sector B,
                         physical closure, T6, or T7.
Current package status:  Exact v1.2 operator-confirmed entry; bundle-last
                         publication authorized after final validation.
Next required action:    Build, independently integrity-check, and register
                         the A5 T5 PUBLISHED capstone bundle last.
```